

Deep Dive: 2. Variables & Data Types

Concept Summary

Variables are named storage locations in memory used to hold data. Python is dynamically typed, meaning it automatically figures out the data type.

Coding Structure

```
variable_name = value
```

Example Code

```
age = 25
price = 19.99
name = 'Alice'
is_active = True
```

Code Explanation

- Line 1: Creates an Integer (whole number) variable named 'age'.
- Line 2: Creates a Float (decimal) variable named 'price'.
- Line 3: Creates a String (text) variable named 'name'.
- Line 4: Creates a Boolean (True/False) variable named 'is_active'. Notice that 'True' is capitalized.

Practice Problems

1. Create variables for your favorite color, your age, and whether you like coffee (boolean). Print them all.
2. Reassign the age variable to be a string ('twenty') and print it to prove Python is dynamically typed.